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Streamflow and Surface-Water Presence Data Availability Across the Conterminous United States: A Review for Headwater Systems

Roy Sando[1] | Kristin L. Jaeger[2] | Christa A. Kelleher[3] | John Hammond[4] | Jay R. Christensen[5] | Catalina Segura[6] | Heather E. Golden[5] | Frederick Cheng[7] | Admin Husic[8] | C. Nathan Jones[9] | Charles R. Lane[10] | Li Li[11] | D. Tyler Mahoney[12] | Hilary K. McMillan[13] | Adam N. Price[6,14] | Erin C. Seybold[15,16] | Adam S. Ward[17] | Margaret Zimmer[18] | Steven Pestana[19]

[1]U.S. Geological Survey, Wyoming-Montana Water Science Center, Helena, Montana, USA | [2]U.S. Geological Survey, Washington Water Science Center, Bellingham, Washington, USA | [3]Department of Civil and Environmental Engineering, Lafayette University, Easton, Pennsylvania, USA | [4]U.S. Geological Survey, Maryland-Delaware-DC Water Science Center, Catonsville, Maryland, USA | [5]Office of Research and Development, U.S. Environmental Protection Agency (EPA), Cincinnati, Ohio, USA | [6]Department of Forest Engineering, Resources & Management, Oregon State University, Corvallis, Oregon, USA | [7]Department of Environmental Sciences, University of Virginia, Charlottesville, Virginia, USA | [8]Department of Civil and Environmental Engineering, Virginia Tech, Blacksburg, Virginia, USA | [9]Department of Biological Sciences, University of Alabama, Tuscaloosa, Alabama, USA | [10]Office of Research and Development, U.S. Environmental Protection Agency (EPA), Athens, Georgia, USA | [11]Department of Civil and Environmental Engineering, Pennsylvania State University, University Park, Pennsylvania, USA | [12]Department of Civil and Environmental Engineering, University of Louisville, Louisville, Kentucky, USA | [13]Department of Geography, San Diego State University, San Diego, California, USA | [14]US Department of Agriculture, Pacific Northwest Research Station, U.S. Forest Service, La Grande, Oregon, USA | [15]Kansas Geological Survey, University of Kansas, Lawrence, Kansas, USA | [16]Department of Geology, University of Kansas, Lawrence, Kansas, USA | [17]Department of Biological & Ecological Engineering, Oregon State University, Corvallis, Oregon, USA | [18]Department of Soil and Environmental Sciences, University of Wisconsin, Madison, Wisconsin, USA | [19]TealWaters, Seattle, Washington, USA

Correspondence: Roy Sando (tsando@usgs.gov)

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ABSTRACT

Water is essential for life on Earth, supporting ecosystems, human health, and economic activities. Hydrology relies on observational data, and this paper discusses regional and national datasets for the conterminous United States (CONUS) publicly available as of 2023, focusing on headwaters, defined as first- and second-order streams at 1:24000 scale. It identifies 72 primary and secondary datasets and 11 repositories and argues how better integration and accessibility of hydrological data can improve research. The paper distinguishes between datasets where streamflow was the primary data collection objective and those where it was secondary. This distinction highlights opportunities to consider data from efforts peripheral to hydrology but is still useful for understanding hydrologic conditions. The analysis reveals that out of about 118000 active and inactive stream observation sites, about 6.6% and 25% are located on first- and second-order streams, respectively. This indicates a substantial data gap for headwater systems, which account for over 77% of stream length in CONUS. Federal agencies manage 72% of hydrologic monitoring sites across all stream orders, but only 34% of these are in headwater systems. Academic institutions operate about 2% of sites, with almost half (48%) in headwater systems, focusing on ecosystem research. State agencies also operate about 2% of sites, primarily on larger systems, with 19% on headwaters. Additionally, 23% of sites are managed by multiple agencies. Spatial patterns further reveal pronounced disparities among physiographic regions. Eastern and coastal provinces show relatively dense monitoring, while central and western regions show sparse coverage. These gaps reflect historical priorities, logistical

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constraints, funding limitations, and the high cost of continuous instrumentation. To address biases in monitoring networks, data collection could be enhanced with low-cost monitoring, community science, and remote sensing technologies. This study also notes the benefits of long-term monitoring and prioritizing retention of streamgages with longer records.

1 | Introduction

Water is fundamental to life on Earth. Societies have followed, philosophized, studied, monitored, and modelled water for thousands of years. The foundations for the first recorded theories about hydrological phenomena can be traced back to the ancient Greeks and Romans (Koutsoyiannis and Mamassis 2021). Despite these early foundations, many misconceptions in hydrology were attributed to the blind reliance on reason (as opposed to empirical observation) exhibited by early Greek and Roman philosophers (Koutsoyiannis and Mamassis 2021). It was not until the 1500s that the formal field of hydrology gained its footing in science, and the importance of reliable observation was established (Rosbjerg and Rodda 2019). Leonardo da Vinci is credited with the earliest published streamflow velocity measurements in his book *Del moto e misura dell'acqua*, written around 1500 but published much later (da Vinci 1828; Koutsoyiannis and Mamassis 2021). In the intervening centuries, water-resources management has been greatly influenced by the volume and variety of observational hydrologic data.

Most modern hydrologic modelling and understanding depend on observational data (Silberstein 2006). Hydrologic data across varying temporal (e.g., discrete, hourly, daily, seasonal, annual) and spatial (local, reach, catchment, regional) scales are used to understand hydrologic mechanisms and predict hydrologic dynamics (Blöschl and Sivapalan 1995; Clark et al. 2015). As outlined by the Community Coordinating Group on Integrated Hydro-Terrestrial Modelling (Scheibe and Stafford 2020), hydrological studies across scales are important for understanding water availability for consumption (Moore et al. 2015), ecosystems (Chang and Bonnette 2016), managing water resources (McDonnell 2008), assessing climate change impacts on streamflow and water quality (Chawla and Mujumdar 2015), and forecasting and mitigating water-related disasters (Merz et al. 2020). Research at the fine scale (e.g., hillslope, reach) can be used to identify specific hydrologic processes and landscape configurations that exist in different systems (Burt and McDonnell 2015; Blume et al. 2017). Additionally, local-scale understanding can help researchers assess how hydrologic processes are influenced by site-specific factors such as soils, vegetation, topography, and disturbance. Conversely, there is also a need for understanding and prediction across larger geographic extents, from regional to global scales (Blöschl 2006; Archfield et al. 2015).

As new and recurring hydrologic and water management challenges arise, scientists can explore and revisit observational data to create, amend, and validate existing hydrologic theories (McCabe et al. 2017). The rate at which these theoretical advancements occur is dependent, at least in part, on the awareness and understanding of existing data. National, regional, and finer-scale hydrologic datasets date back to the 1800s in the United States (Lanfear and Hirsch 1999). We are now in an age of data generation, with the advent of large, geographically extensive datasets that leverage remotely sensed data and in situ observations at increasingly finer temporal resolutions and greater accuracy (Revilla-Romero et al. 2016). Specifically, there has been substantial advancement in the field of large-sample hydrology through the global expansion of robust and standardized datasets (Addor et al. 2020). Assembling these datasets, online links, and descriptions can aid in the efficient findability and use of data and help identify data gaps. Integrating new databases with older, historical datasets can maximize our use of these data and, ultimately, our understanding of hydrology and hydrologic systems.

While the occurrence of cumulative, publicly available global data and standards is increasing, there is also a bias in when and where hydrological data are collected along the river network. In particular, headwaters, the most upstream extent of the channel network and defined here as Strahler stream orders 1 and 2 following Golden et al. 2025, are typically under-monitored compared to their dominating prevalence in the landscape (Deweber et al. 2014; van Meerveld et al. 2020; Krabbenhoft et al. 2022). In the United States, headwater monitoring is primarily conducted at a handful of intensively monitored sites. These data can improve modelling efforts because they often represent the greatest variety of hydrologic conditions. Headwaters account for most of the global channel network by length (Allen and Pavelsky 2018; Lane et al. 2026) and provide critical hydrological, biogeochemical, and ecological functions across watershed scales (Battin et al. 2008; Meyer et al. 2007; Wohl 2017). However, headwater datasets are often not included in national repositories, making their discovery more challenging and limiting synthesis across sites.

The objectives of this paper are to inventory and characterize streamflow datasets available across the conterminous United States (CONUS) through 2023 Specifically, the questions we ask are (A) “What streamflow and surface-water presence datasets exist across CONUS, and how do their spatial and temporal observational characteristics vary across monitoring networks?”; (B) “How well do those observational networks represent headwater streams compared to downstream systems?”; (C) “How does monitoring coverage vary across physiographic regions of CONUS?”, and (D) “What systemic biases, logistical constraints, motivational, and institutional patterns influence where and how hydrologic monitoring occurs across CONUS?” To the authors' knowledge, this is the first assemblage and characterization of streamflow data at this scale in the CONUS. We focus on datasets with spatial domains that are state, regional, or national, or that are part of a monitoring network spanning regional or national extents. We distinguish between datasets in which streamflow was the primary data collection objective and those in which streamflow is a secondary measurement to primary data (e.g., biological, ecological). Although secondary datasets are auxiliary and have less temporal and spatial coverage, they still provide information on the presence or volume of streamflow. By distinguishing between these two types of data collection, we provide insight into just how much additional data can be obtained by inclusion of non-traditional

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streamflow data. We provide descriptions of the temporal and spatial characteristics of the datasets, and include an assessment of the representation of headwater systems in the current state of hydrologic monitoring efforts. Because of the increased recognition of the importance of headwaters (Golden et al. 2025), this work could be particularly important for advancing our understanding of hydrological processes in headwater systems. Though our focus is on CONUS, the approach we use here can be adapted to other countries and regions around the globe.

2 | Methods

2.1 | Data Collection

A combination of methods was used to identify and describe existing streamflow datasets. We first identified publicly available datasets known and used by the collective authorship team, and second, developed key words and terms for a Google engine search. The first phrases used included "hydrologic observation dataset" and "hydrologic monitoring". To identify state-specific datasets, we used the same keywords with the addition of each state name. For example, a search for datasets in Montana would have included the keywords "hydrologic observation dataset Montana" and "hydrologic monitoring Montana". While we attempt to represent a comprehensive set of existing, publicly accessible datasets, other datasets may exist that are not included here. In addition, the list of aggregated datasets does not include one-off, individual projects (e.g., university-led, short-term research efforts) or datasets that were collected with the objective of understanding a single watershed (e.g., the Clark Fork coalition in Montana; https://clarkfork.org). Instead, datasets that are available at the state, regional, or national level were the focus. However, exceptions to this rule occurred when datasets were included as part of a previous assemblage (e.g., CHOSEN; Zhang et al. 2021). While it was outside the scope of this work to process new data from water quality studies or datasets, these sources can be mined and processed similar to Sando et al. (2022).

2.2 | Data Characterization

For each dataset, we gathered characteristics of the data, including the network it is associated with, data objective type ("Primary", "Secondary"), objective frequency type ("Continuous", "Discrete"), total sites, state location, temporal extent, temporal resolution, objective (e.g., reason for data collection), responsible agency, and others (Tables S1 and S2). We summarize data collection efforts based on objective and objective frequency type. Primary datasets contain data collected to record repeated, direct observations of the volume or presence of streamflow at a given site with a unique set of coordinates. These primary datasets are often characterized as streamflow monitoring networks. Secondary datasets represent data that were collected with a primary objective other than recording streamflow but still contain information on the presence or volume of streamflow at a site. Often, secondary datasets contain discrete observations and therefore might require more processing to extract information about water volume, rate, or presence. However, because secondary data are typically collected as part of a larger study, they are usually paired with other, non-hydrological data, which can add value in terms of interpretation or modelling (Salmon-Monviola et al. 2025). Examples include aquatic habitat surveys or riparian condition assessments. We categorized datasets as discrete if there were not consecutive daily observations over the course of at least 3 months; otherwise, they were labelled continuous.

Datasets were collected and published by many entities including federal agencies such as the U.S. Geological Survey (USGS) and the US Forest Service (USFS), state agencies, typically related to Departments of Natural or Water Resources, academic institutions, and other organizations. The responsible agency manages all aspects of data collection and dissemination including funding operation and maintenance costs of data collection, data collection procedures, quality assurance, metadata documentation, and data dissemination. The extent and character of the data will depend on the capacity of the responsible agency and can be highly variable, with larger federal agencies often having more resources to maintain robust, standardized, and geographically extensive data collection efforts (Kaiser, Blasch, and Schmitz 2023). We assigned stewardship according to the number of monitoring sites and associated years included in datasets that were associated with an agency or institution. We assigned a stewardship of 'other' to sites that were included in datasets that were a collaboration among multiple agencies and (or) entities. If a site was published by multiple agencies for non-overlapping periods, we attributed the respective steward to each period. For example, if a site was operated by the USGS from 1901 to 1941, but operated by the state of Nebraska from 1941–2000, we assigned the responsible agency as "Federal" from 1901 to 1941 and "State" from 1941 to 2000. We also collected information on each site, including geographic coordinates of the observation, associated stream order (according to the National Hydrography Dataset Plus High-Resolution; NHD+ HR; USGS 2019), active years of operation, and specific start and end years. These associated data were obtained by identifying the NHD+ HR catchment that the site is located in and using the Unique Identifier value to identify the basin and associated stream reach. Headwater status was assigned to a site if it was located in a catchment that is associated with a first- or second-order stream; if not, it was defined as a downstream site. We also recorded the repository that contained each dataset we include in this analysis (Table S3). The repository that we obtained the dataset from is listed as the primary repository, but the datasets might be contained in multiple repositories.

2.3 | Physiographic Regions

Stream gages have an acknowledged global bias in placement, with bigger and more populated watersheds being more heavily monitored (Krabbenhoft et al. 2022). Because of this, there is likely an unbalanced representation of landscape conditions being monitored. To explore this, we compared the compiled monitoring data from the identified datasets, represented by total days of observations and summarized them for headwater and downstream regions. Total days of observations were normalized by the total length of Strahler order 1 and 2 (headwaters, collectively), and Strahler order 3 and higher (downstream) across 24 physiographic provinces as identified by Fenneman and Johnson (1946). We opted for physiographic provinces to describe the CONUS to leverage spatial continuity and geographic cohesion.

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Streamgage-representation studies that discretize by hydrologic unit codes (HUCs; e.g., Van Metre et al. 2020) are bounded by watersheds that often encompass different physiographies and climates. Hydrological classification (e.g., Wolock et al. 2004) results in discontinuous regions and numerous locations, making evaluation and interpretation difficult. The physiographic provinces presented a reasonable middle ground, as they generally describe the CONUS, result in contiguous subregions, and offer a more readily interpretable overall number of regions. The site of the streamgage or observation location, rather than the basin location, was used for designating physiographic provinces. To assess the representation of headwaters in the monitoring network relative to the proportion of headwater streams in the hydrologic network of each physiographic region, we summed the total length of NHD + HR Strahler stream order 1 and 2 (headwaters) and Strahler stream order 3 and greater (downstream).

3 | Results

3.1 | Streamflow Data, Monitoring Networks, and Collection Objectives

As of 2023, 72 historical and ongoing primary and secondary datasets were identified at about 118 000 sites in the CONUS (Figure 1; Table S1; Sando et al. 2026). These datasets range in size, spatial, and temporal extent, with the largest and longest dataset being the U.S. Geological Survey's National Water Information System (NWIS; U.S. Geological Survey 2024) that contains about 32 000 sites dating back to 1857. The size of all other datasets ranges from 1 to about 28 000 sites with a median of 10 sites per dataset. Periods of record for non-USGS datasets range from 1900 to 2023, though collection for many of these datasets is ongoing.

The organizations that have supported collection and cataloguing of these datasets exist for objectives that are broadly organized into six themes (Table 1). The earliest efforts to monitor U.S. water resources date back to the late 1800s and were intent on understanding resources available for consumptive use. For example, the oldest and most extensive network is the USGS' stream gaging network, served through the National Water Information System (NWIS; U.S. Geological Survey 2024), which was originally implemented to primarily understand volume and regularity of discharge for the purposes of irrigation and water use (Rabbitt 1975). Today, these data are used in a multitude of studies covering a vast array of scientific topics (e.g., Cole et al. 2018; Juracek and Fitzpatrick 2009; Walker et al. 2020). The National Aquatic Resource Surveys (NARS) contain discrete data and were created by the U.S. Environmental Protection Agency to assess the quality and condition of the nation's water using a probabilistic survey. NARS-based surveys are conducted on rivers and streams once every 4 years; wetlands and lakes data are collected on similar four-to-five-year cycles but in different years (U.S. Environmental Protection Agency 2006).

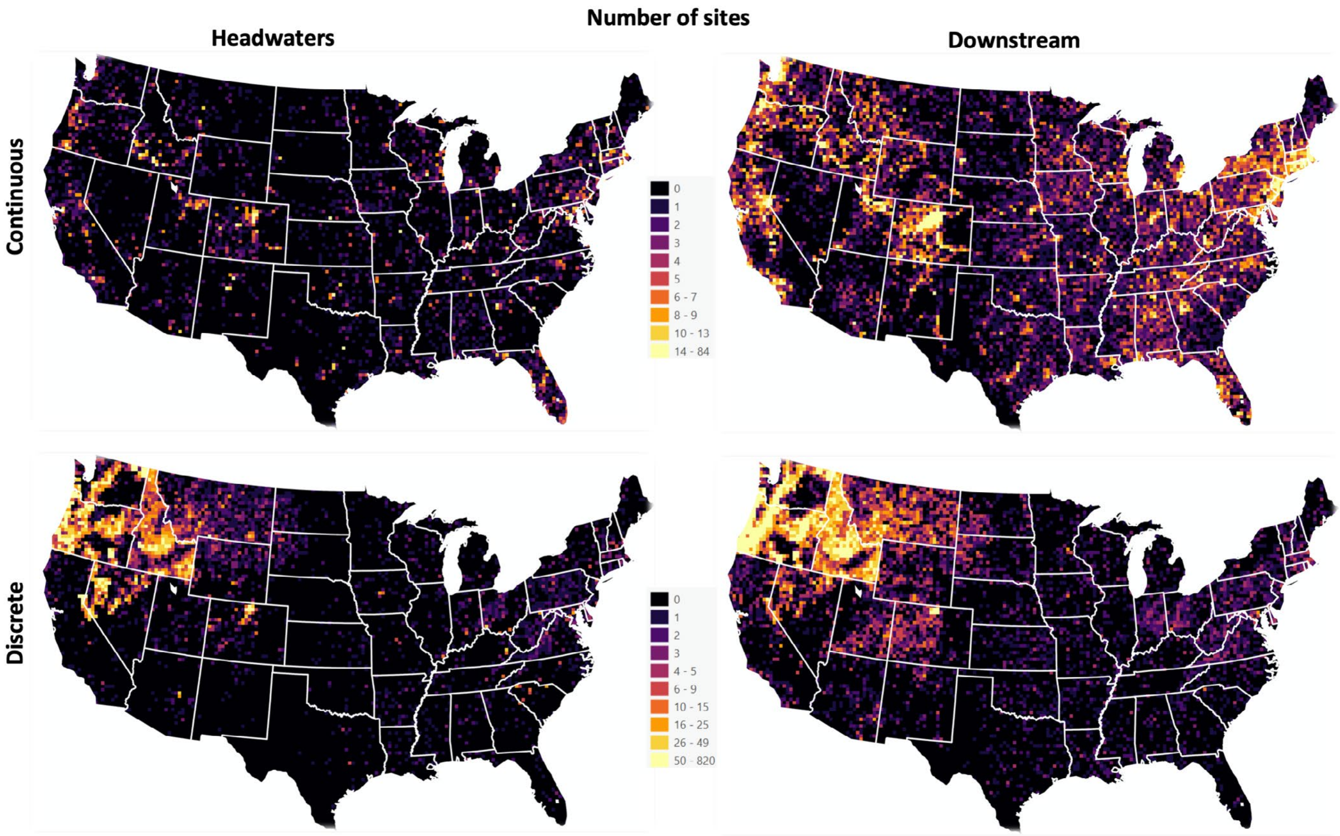

FIGURE 1 | Number of continuous and discrete streamflow monitoring sites classified by headwaters (Strahler order one or two at 1:24000 scale) versus downstream (Strahler order 3+) for a 25 × 25 km grid across the conterminous United States from the efforts identified in this study (Sando et al. 2026).

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TABLE 1 | Seventy-two streamflow datasets that were used in the analyses and characterizations described in this manuscript categorized into objective focus with the number and proportion of headwater sites associated with each theme.

Theme	Objective focus	Datasets	Headwater sites (percent of sites for theme)
1. Streamflow monitoring and water resource management	Continuous monitoring to support water allocation, flood forecasting, and water availability assessments	USGS Streamgages, State streamgaging programs (Massachusetts, Minnesota, Montana, Colorado, Oregon, Utah, Washington, Nebraska, New Hampshire), Upper Arkansas Water Conservancy District, Northern Colorado Water Conservancy District, NEON, Streamflow Duration Assessment Method observations, National Aquatic Resource Surveys, CrowdWater, StreamTracker, FLOwPER, Streamflow Observation Points (Upper Missouri, Pacific Northwest), Dataset for Streamflow Data Catalogue for the Pacific Northwest, Flow Photo Explorer, EPA Headwaters	36 260 (31.5%)
2. Forest hydrology and watershed management	Understanding forest-water interactions, sedimentation, and effects of forest management	Experimental forests (Caspar Creek, Tenderfoot, Fraser, San Dimas, Hill Demonstration, Boise Basin, Coram, Fernow, Glacier Lakes, Priest River, Santee, Marcell), Aquatic Inventory and Monitoring, PacFish Infish and Biological Opinion, Walnut Gulch, Steamboat Creek	531 (21.8%)
3. Agricultural impacts and land use practices	Evaluating agricultural practices, soil health, nutrient transport, and watershed-scale management	Longterm agricultural research stations (Central Mississippi, Eastern Corn Belt, Great Basin, Gulf Atlantic Coastal Plain, Lower Mississippi, Southern Plains, Texas Gulf, Upper Chesapeake Bay), Walker Branch, ARS Experimental Watersheds	254 (74%)
4. Critical zone and biogeochemical processes	Investigating interactions among water, soil, vegetation, and geochemistry in the Critical Zone	Critical zone observatories (Boulder Creek, Calhoun, Christina River Basin, Intensely Managed Landscapes, Santa Catalina/Jemez, Southern Sierra, Susquehanna Shale Hills), East River Scientific Focus Area, Bear Brook	26 (48.1%)
5. Urban hydrology, socio-ecological systems	Studying hydrologic processes in urban environments, human-environment interactions, and community science	Baltimore Ecosystem Study, Central Arizona	4 (30.8%)

(Continues)

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TABLE 1 | (Continued)

Theme	Objective focus	Datasets	Headwater sites (percent of sites for theme)
6. Long-term ecological research (LTER) and ecosystem dynamics	Long-term monitoring of ecological and hydrological processes across diverse ecosystems	Coweeta, Florida Coastal Everglades, Andrews, Harvard Forest, Hubbard Brook, Jornada Basin, Konza Prairie, Niwot Ridge, Plum Island Ecosystems, Santa Barbara Coastal, South Umpqua, Virginia Coast Reserve	44 (65.7%)

Note: Dataset citations are provided in Table S1.

All states provide funding to the USGS to support streamflow monitoring in their respective watersheds. Nine states were identified that also operate a state-specific streamflow monitoring network, or have datasets published and served by state government agencies, to supplement the USGS streamgaging network (Table S1). Some states monitor diversions (e.g., Idaho Department of Water Resources) or lake levels (e.g., Oklahoma Water Resources Board) and many states beyond the nine identified here monitor groundwater wells. Diversions, lake, and well monitoring are not included in this analysis, given the streamflow focus. Additionally, virtually all states operate an ambient water quality monitoring program, but harvesting and processing these data into streamflow information can take a substantial effort. Finally, many states may have individual cities or counties that operate a streamgaging network. Networks operated by individual cities and counties are not included in this analysis. Exceptions to this rule are networks included as part of a larger assemblage of data (e.g., CHOSEN; Zhang et al. 2021). Collecting datasets from cities, counties, and individual watersheds was outside the scope of this work.

Some networks have been strategically implemented to monitor hydrological processes as they relate to ecosystem or earth system science understandings, such as hillslope and critical zone processes and impacts of different land uses or disturbance histories such as agriculture (Long Term Agroecosystem Research [LTAR] stations), timber harvest (Andrews Experimental Forest), or other intensely managed landscapes (e.g., Intensely Managed Landscapes Critical Zone Observatory). The National Science Foundation's Long-term Ecological Research (LTER) Network was created with the intent of focusing on ecological questions that require time scales of years, decades, and centuries to address basic science questions (Hobbie et al. 2003). Streamflow monitoring at LTERs have been used in a myriad of multidisciplinary studies of impacts on streamflow and watershed hydrology some of which include management applications, including the role of future land cover and climate (Barnhart et al. 2021), forest management practices (Perry and Jones 2017), and other ecosystem processes and human influences (Jones et al. 2012). Critical Zone Observatories (CZOs) are a more recent effort aimed at collecting data to improve understanding of earth system sciences. CZOs have existed since 2007 (Lin et al. 2011) and were initially funded by the U.S. National Science Foundation. Many of these data were compiled in the Comprehensive Hydrologic Observatory Sensor Network (CHOSEN) database, which was developed for the purpose of addressing the dearth of long-term, regionally extensive, hydrologically comprehensive databases (Zhang et al. 2021).

Other networks were initiated with the goal of monitoring and assessing land use and conditions of public lands to more directly inform management decisions. Data associated with the U.S. Bureau of Land Management's Aquatic Inventory and Monitoring (AIM; Taylor et al. 2014) protocol were collected to monitor the condition and trend of aquatic ecosystems (Bureau of Land Management 2021). The U.S. Forest Service's experimental forests (EFs; Adams et al. 2004) are ecologically focused with monitoring to improve understanding of catchment hydrology and ecosystem processes (e.g., San Dimas EF) and land use change, notably related to timber harvest and restoration (e.g., Tenderfoot EF, Priest River EF, Caspar Creek EF, Fernow

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EF). The origins of the EF network date back to the early 1900s, with the earliest streamflow monitoring dating back to 1939 (Table S1).

Many of these networks were created to monitor and improve understanding of impacts associated with human activities on river systems and aquatic ecosystems including some CZO (e.g., Kumar et al. 2020), LTER (Baltimore; Cary Institute of Ecosystem Studies and Band 2013), and Experimental Watersheds and their successor program, the Long-term Agricultural Research Networks (LTAR; e.g., Reba et al. 2011). The Agricultural Research Service (ARS) Experimental Watersheds Program (Goodrich et al. 2021) advances agricultural and environmental research by studying interactions between agricultural practices, hydrological processes, and ecological systems. By providing measurements of streamflow, these observations establish baseline conditions that serve as a reference for assessing changes over time.

Recent initiatives, such as FLOwPer, CrowdWater, Crowd Hydrology, and StreamTracker, which were all considered as part of this work, incorporate the public in collecting hydrologic data as a low-cost expansion into smaller systems (Jaeger et al. 2020; Kampf et al. 2018; Seibert et al. 2019). Community science is being recognized globally as a viable alternative to installation of continuous stream monitoring equipment (e.g., Alemu et al. 2023). In the CONUS, these crowd-sourced science initiatives have primarily focused on collecting simple visual observations of surface water presence or absence through the use of smartphone apps. As part of targeted sampling campaigns or on their own time, participants record if surface water is visibly present in a stream, as well as other metadata such as coordinates, date and time of record, geomorphologic conditions, and other information. Outside of targeted campaigns, the data may be spatially and temporally independent. These data are discrete (non-continuous) visual observation of surface water presence or water level at a stream location that can be collected at many locations (tens of thousands of locations) across large geographic extents as opposed to intensive monitoring with instrumentation at fewer locations (Truchy et al. 2023). These datasets have been used to train stream permanence and flow models (e.g., Avellaneda et al. 2020; Jaeger et al. 2019; Peterson et al. 2024; Sando et al. 2022). These new methods for monitoring streamflow data are included as part of this assessment.

3.2 | Data Collection Biases and Challenges

3.2.1 | Headwater Stream Monitoring

Most hydrologic monitoring to date has focused on larger, downstream systems (i.e., stream orders greater than 2). Conversely, we found that headwater streams are severely underrepresented in publicly available datasets relative to proportional abundance in river network systems (Figure 2). Out of a total of 117 903 active and inactive stream observation sites, 7782 (about 6.6%) and 29 688 (about 25%) sites were in Strahler first- and second-order catchments, respectively. This disparity is further emphasized when we examine the number of observation days associated with headwater streams. There were about 322 million streamflow observation days in the CONUS. Of these, only about 11.7 million days (about 3.6%) were observed on first-order streams, and about 27.2 million days (about 8.4%) on second-order streams. However, assessment of the total length of streams in the NHD + HR dataset revealed that about 56% and 21% of stream length in CONUS is associated with Strahler stream order 1 and 2, respectively. If the proportion of observation sites would reflect the proportions of the channel network that the observations represent, 77% of the observational dataset should be in headwaters (instead of only 31%). This work highlights that, even when accounting for a broad assemblage of disparate datasets, headwaters remain underrepresented in terms of streamflow data.

3.2.2 | Physiographic Representation

Our results show that physiographic provinces in the east and along the west coast of the CONUS have the greatest representation in the hydrologic monitoring network (Figure 3A). Conversely, physiographic provinces in the central and western United States have the lowest density of observation days. The central and western United States have lower population densities than coastal regions but have undergone intensive land management practices, including decades of agricultural and grazing practices. These land-use practices can have profound implications on hydrological processes (Kim et al. 2023). Additionally, water availability remains a persistent issue for the central and western United States (Ketchum et al. 2023). The southern Rocky Mountains (J.) are a notable outlier in the otherwise sparse data regions of the central and western United States. A potential explanation for this is the importance of the Colorado River basin as a water source for the western United States. The Colorado River is one of the most critically important and heavily managed river systems in the world, supporting the water needs of more than 40 million people, extensive agricultural production, hydropower generation, Tribal communities, and internationally shared water allocations (Tillman et al. 2022). We also observe that the relative proportion of headwater monitoring is consistently low for all physiographic regions, with a mean and standard deviation of 6.93 and 6.91 observation days per kilometre, respectively (Figure 3B). Conversely, on downstream systems, there is a mean and standard deviation of 132 and 53.5 observation days per square kilometre, respectively.

3.2.3 | Long-Term Monitoring

There is a lack of continuous long-term monitoring data in most watersheds, particularly in areas where there may be limited funding for research and monitoring (Moran et al. 2008). Our analysis shows that only about 2% of continuous headwater (Strahler stream orders 1 and 2) monitoring sites and about 10% of continuous downstream sites had periods of record of 100 years or longer (Figure 4). Thus, there are about 3700 continuous monitoring sites in the CONUS with more than 100 years of record. This comes out to a mean density of about one site per 2200 square km. Further, our analysis shows that about 28% and 63% of continuous downstream sites had periods of record with 50–99 years and less than 50 years, respectively. Conversely, about 14% and 84% of continuous headwater sites had periods of record with 50–99 years and

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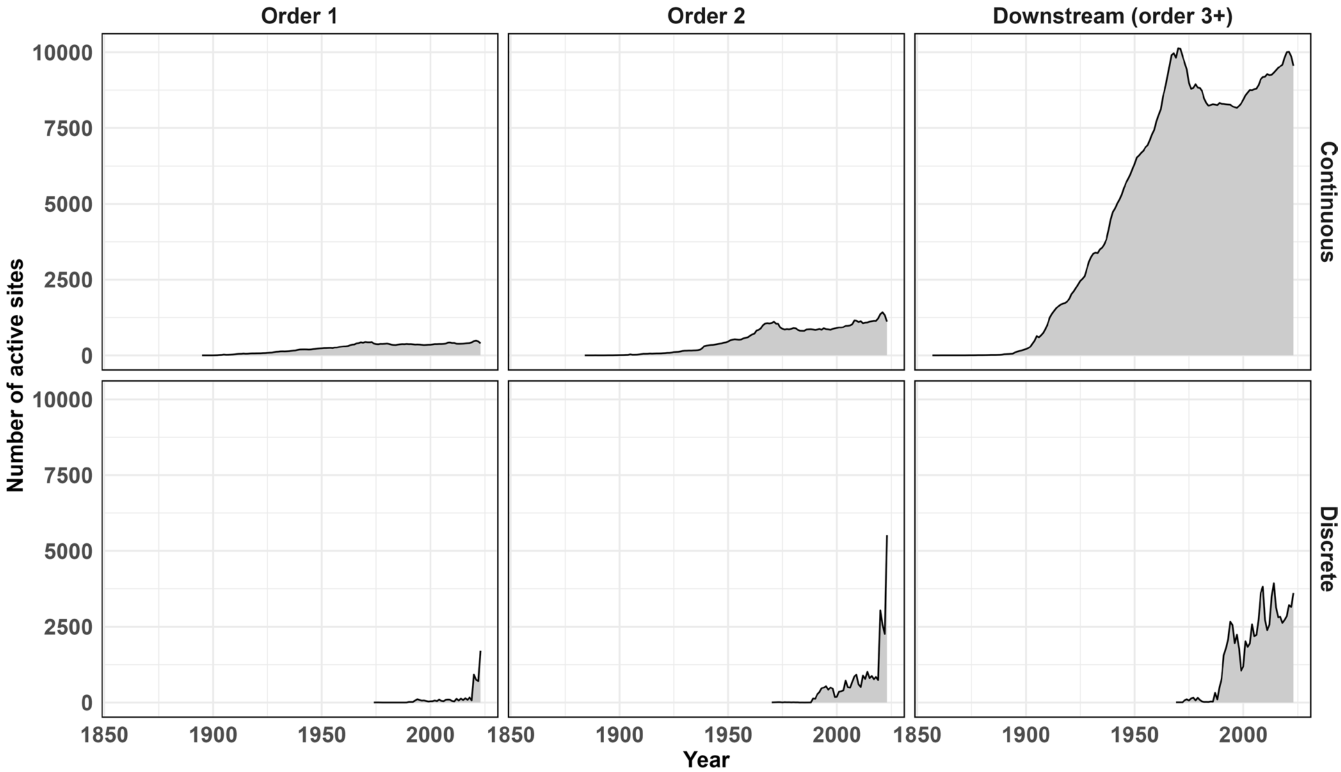

FIGURE 2 | Total annual number of continuous and discrete hydrologic monitoring sites for headwater streams (Strahler stream order one or two) and downstream systems (Strahler stream order 3+) in the Conterminous United States (1857–2023).

less than 50 years, respectively. These results also show that larger systems tend to have a greater percentage of sites with longer periods of record (i.e., greater than 50 years), emphasizing the bias toward larger systems that are managed for their water resources.

3.3 | Summary of Data by Responsible Agency

Many agencies collect data on various aspects of U.S. streams and rivers. At the national scale of this study, the Federal government is by far the largest steward of hydrologic observation data. Collectively, federal datasets account for about 72% of the total hydrologic monitoring data across the CONUS (Figure 5A) underlining the critical role that these agencies play in natural resources management and protection. However, because the priority for these efforts is often understanding aspects of hydrology that are better captured by monitoring larger, downstream systems (Strahler stream orders 3 and greater) (Wahl et al. 1995), only about 34% of Federal data are in headwater systems, or catchments associated with Strahler stream orders 1 and 2 when mapped at a 1:24000 resolution (Golden et al. 2025). Larger scale and older datasets (e.g., National Aquatic Resource Surveys) were designed on coarser hydrographic networks (e.g., 1:100000) in which Strahler stream orders 1 and 2 may correspond to larger streams or rivers that would be identified as higher order streams in higher resolution datasets. Because of this legacy design on coarser resolution, datasets with streams mapped as headwaters at cartographic resolution of 1:24000 are likely underrepresented. Further, even 1:24000 resolution may not be capturing true first- and second- order streams (Botter et al. 2026), further emphasizing the bias in headwater representation of federal datasets.

The second largest category of sites is associated with multiple organizations and classified as 'Other'. These account for about 24% of sites and, like Federal sites, are heavily biased toward downstream sites with about 77% in Strahler order 3 or greater (Figure 5E). Similar to 'Federal' and 'Other' sites, data published by state agencies have an almost exclusive focus on continuous monitoring of larger, downstream systems. Almost 81% of these data are located on streams of Strahler order 3 or greater (Figure 5D). However, state-published datasets only account for about 2% of total streamflow data in the CONUS. Conversely, the next largest category of sites is collected by academic institutions. These also account for only about 2% of sites, but a greater proportion of sites in headwater systems (52%) and a majority of data that are discrete (97%) (Figure 5B,C). Conversely, about two-thirds of federally collected data are discrete. These differences in data collected by academic institutions versus other organizations are presumably a consequence of limited long-term (greater than 5 years) institutional funding.

3.4 | Data Services and Repositories

Eleven data services or repositories for hydrologic monitoring data were identified at the national or regional (multi-state) scale (Table S3). Some of these repositories have been built for a single, specific dataset, with a common set of standards for those data

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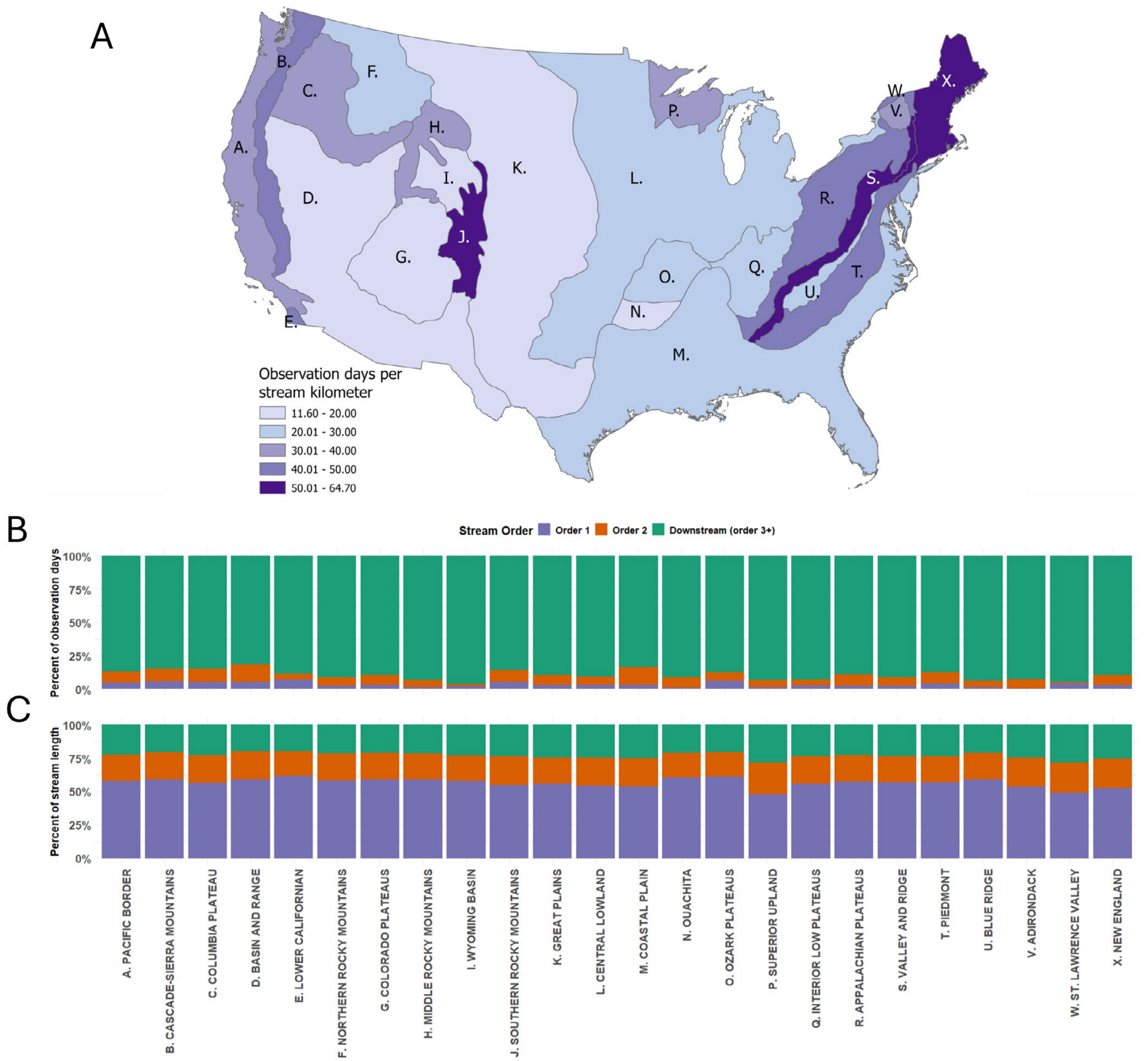

FIGURE 3 | Hydrologic observation days per stream kilometre for each physiographic region in the study area shown spatially (A) and proportionally by stream order (B). We also provide the proportion of headwater streams (Strahler stream order one or two) and downstream rivers (Strahler stream order 3+) (C). Physiographic regions (Fenneman and Johnson 1946) are listed from west to east in the bar plots (B and C).

(e.g., NWIS; U.S. Geological Survey 2024). Similarly, the online database Sustaining the Earth's Watersheds—Agricultural Research Data System (STEWARDS; Sadler et al. 2020) was built by the USDA's Agricultural Research Service to deliver weather, hydrology, and water quality data to the public and contains data from the Long-term Agricultural Research Stations (LTARs).

Other efforts have focused on pulling together thematically similar data for the purpose of providing more information on a particular branch of science. For instance, MacroSheds (Vlah et al. 2023) has a primary objective of merging all U.S. federally funded watershed ecosystem studies into a common platform for classification of watershed ecosystems that based on watershed functional traits (*sensu* McDonnell 2008). Others have recognized the need to find and coalesce disparate data that were collected through localized, focused efforts into larger regional datasets that are easier to find and access by other organizations and the public at large (e.g., Kaiser, Blasch, and Hall 2023).

Finally, much effort has been invested into the creation of services designed to provide general capability for serving hydrologic data from a wide variety of studies, agencies, and objectives. For example, HydroShare, curated by The Consortium of Universities for the Advancement of Hydrologic Science Inc. (CUAHSI; https://www.cuahsi.org/), is a web-based hydrologic information system for users to publish hydrologic data in formats following FAIR principles (Wilkinson et al. 2016) and includes web application tools for data analysis by end

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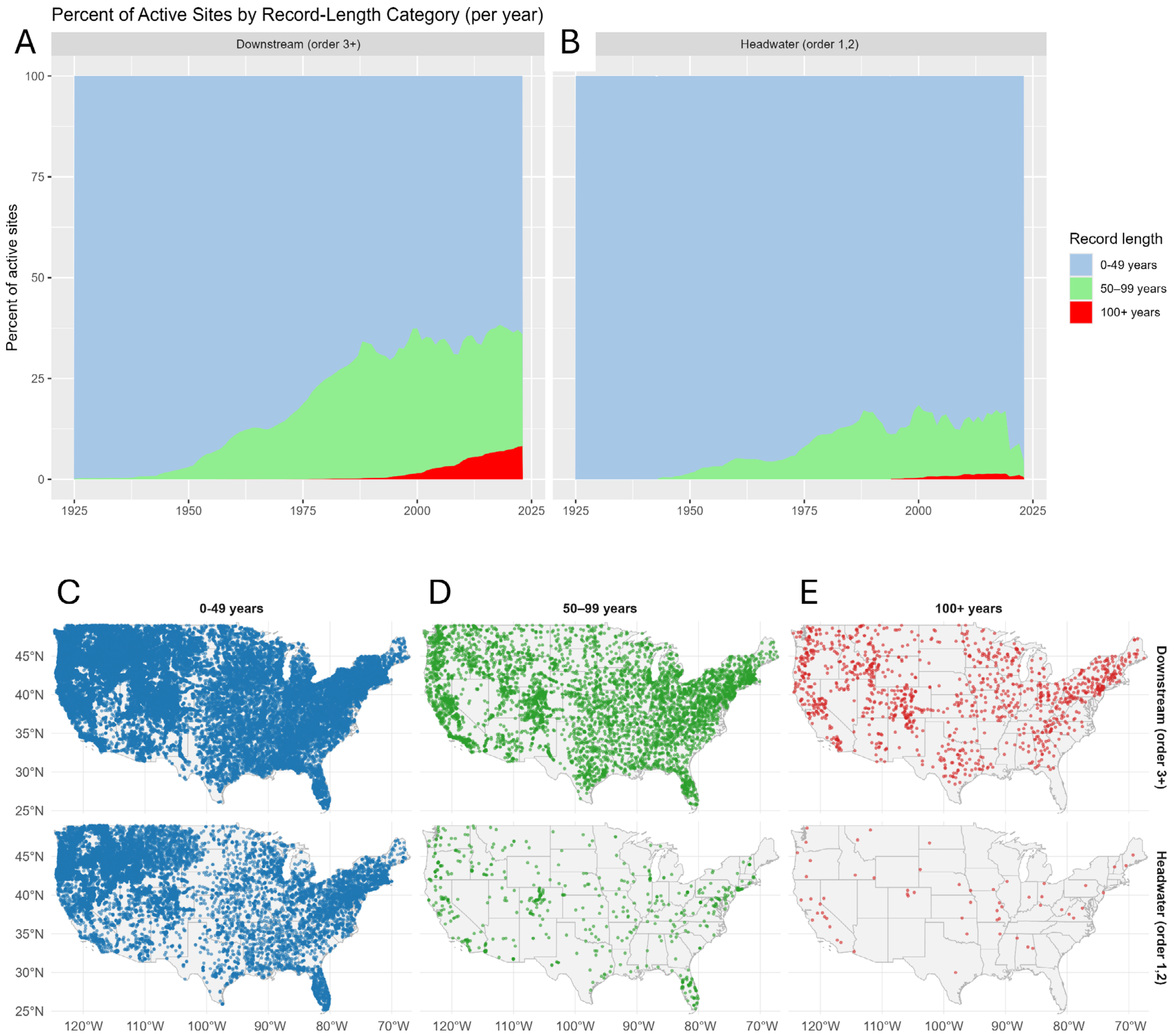

FIGURE 4 | Spatial distribution of continuous headwater (Strahler order 1 and 2 streams at 1:24000 resolution) and downstream (3-order or higher) monitoring sites with (A) less than 50 years of record, (B) 50–100 years of record, and (C) more than 100 years of record. Percent of active continuous monitoring sites by year in headwater (D) and downstream (E) catchments with more than 100, 50–99, and less than 50 years of record.

users. Additionally, the Internet of Water (IoW; https://internetofwater.org/) collectively provide tools to access, publish, and manage hydrologic and related data. Similarly, Estreams (www.estreams.eawag.ch) is another web-based information system that provides access to streamflow, hydro-climatic, and landscape characteristics from several different datasets for Europe.

4 | Discussion

4.1 | Efficient Monitoring for Effective Modelling

Many headwater streams in the CONUS are in remote environments that are challenging to access, such as steep terrain or dense vegetation in the canopy or understory, which make data collection difficult. In addition, the relatively high cost of continuous stream monitoring juxtaposed with the sheer abundance of headwaters remains an ongoing challenge to sufficient representation in hydrologic monitoring networks (Abbott et al. 2018; Golden et al. 2025). In 2008, it was estimated that the annual cost for operation of a single stream gage by USGS ranged from about \$13–\$15 k (Norris et al. 2008), which is about \$19–\$22 k in 2025 after adjusting for inflation. These costs include robust, published protocols on data collection that often include sophisticated instrumentation and technology, post-processing and data review as well as standardized, comprehensive metadata documentation, which is an often-overlooked element of data collection and management (Kaiser, Blasch, and Schmitz 2023), as well as an overall decline in the number of actively reporting gages globally (Ruhi et al. 2018). This level of expense can make it infeasible to fund large, continuous stream gaging efforts. As a result, there currently are substantially fewer data available for

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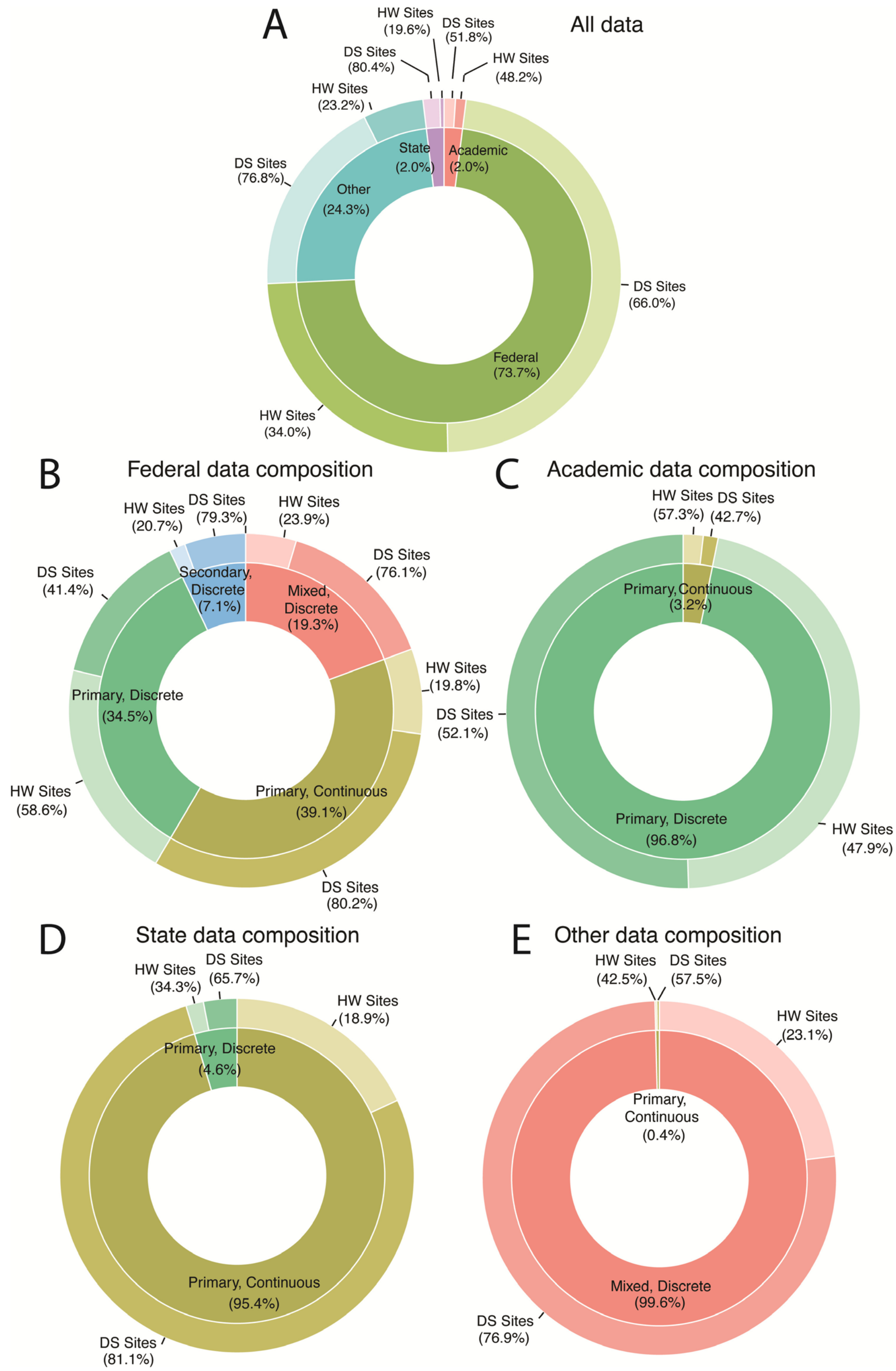

FIGURE 5 | Percentage of data collected by institutions, and the composition of data categories collected by each institution. 'Other' includes nonprofit organizations and collaborations among multiple agencies. Primary datasets are often described as streamflow monitoring networks and have a direct objective of monitoring streamflow. Secondary datasets represent data that were collected with a primary objective other than recording streamflow but still contain information on the presence or volume of streamflow at a site location. Mixed includes both primary and secondary sites. Headwater ("HW") and downstream ("DS") sites are shown as percentages for each dataset or categorization.

headwater streams, limiting our understanding of these systems (Bishop et al. 2008). Beyond the United States, this bias has been identified in global and other national streamgaging networks (Beaufort et al. 2018; Krabbenhoft et al. 2022). Continuous monitoring can range in cost from sensors and techniques that observe water presence or absence (low cost),

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water level or stage only (medium cost), and water level or stage combined with repeat flow measurements to calculate streamflow magnitude (high cost). Thus, expanding the existing streamflow monitoring network through water presence or stage-only monitoring could be achieved at a lower cost, allowing for broader coverage (Chapin et al. 2014; Fritz et al. 2020; Seo et al. 2022). Further, utilizing a variety of monitoring techniques and technology can be used in remote, headwater areas where it is not feasible to install permanent and bulky monitoring equipment (Goodling et al. 2025). To address the immense challenge of estimating the magnitude and timing of streamflow contributions from an adequate number of headwater streams, combinations of (1) additional continuous, low-cost monitoring, (2) repeat discrete measurements at a larger number of sites, and (3) fine-scale airborne and satellite remote sensing can be used. It has been pointed out (Silberstein 2006), as modelling methods advance, the demand for new and better data increases. Following this line of thought, research can be used to improve how precise headwater measurements and predictions can be incorporated into various decision workflows and relevant management strategies.

Expansion of headwater data that leverage cost-effective approaches can lead to more effective modelling and improve predictions. Machine learning methods that can incorporate multiple formats of data have advanced. In addition, spatially based research can provide insights into the temporal patterns of hydrologic systems (Yadav et al. 2007; Singh et al. 2011). Recent work suggested that streamflow duration model performance is optimal when using sites with a minimum of 10 observations per site covering a variety of seasonal flow conditions across an entire 4875 km^2 regional flow network (Peterson et al. 2024). Other research suggests that more traditional methods (e.g., integrated hydrological modelling) are improved by having at least 4 years of continuous daily discharge monitoring (Jiang et al. 2024). Discrete observations have been used to fill in temporal monitoring gaps in streamflow characteristic modelling (Mazzoleni et al. 2015). By collectively leveraging these diverse data types that include spatially distributed discrete observations of streamflow, long-term continuous monitoring at established stream gages, and remotely sensed estimates of flow network expansion and contraction, we could enhance our streamflow estimation capabilities in headwaters (e.g., Vanderhoof et al. 2025).

4.2 | Data Harmonization

Streamflow monitoring networks are implemented to provide information to serve many different interests (Juracek and Fitzpatrick 2009; Stewart 2015). While there is a degree of overlap in the information gained from these disparate efforts, there remain distinct objectives associated with the collection of the different datasets. Different objectives can influence important aspects of the network, including the gaging equipment used, the coordinates of monitoring sites, and the period of record and frequency of observations. Because of these differences, it can be important to harmonize the data from multiple sources to reduce the discrepancy that might be introduced because of these differences. Data harmonization (the process of consolidating, unifying, and standardizing data) in hydrologic modelling is used to ensure consistency, accuracy, and comparability of hydrologic data across various studies and regions (Kaiser, Blasch, and Hall 2023). As hydrologic models rely on diverse datasets, variations in data collection methods, measurement units, and temporal resolutions can lead to discrepancies that make it difficult to develop consistent and interoperable training datasets for data driven and processed-based model applications. Harmonizing data allows researchers and practitioners to integrate information from multiple sources, facilitating more robust analyses and enhancing their suitability and integration across many hydrologic model structures and thereby improve model accuracy. It also adds, and potentially improves, spatial and temporal resolution broadening the applicability and enabling more robust validation of models. Furthermore, standardized data formats and protocols improve collaboration among scientists, policymakers, and stakeholders, enabling more effective water resource management and decision-making (Ward and Packman 2019). In an era of increasing climate variability and water scarcity, harmonized data can improve adaptive strategies that can address complex hydrologic challenges at local, regional, and global scales. It should also be noted that there have been efforts to combine streamflow data with meteorological or other hydro-related phenomena, for example CAMELS (Addor et al. 2017).

4.3 | Data Sharing

Data sharing can improve the advancement of scientific knowledge. Sharing of hydrologic data can reinforce research findings and drive innovation across fields (Yu et al. 2020). Publicly accessible data also facilitate collaboration among researchers and help decision makers to make informed decisions related to natural resources management (Carter et al. 2025). Ongoing data collection, modelling, and synthesis are generating an increasing volume of data and information for hydrological research and management. Without protocols to ensure automatic data discovery, many of these data may become lost as “dark data”, or information collected by organizations that is not actively used or analysed, despite calls for data rescue (Bledsoe et al. 2022; Yu et al. 2020). In addition, syntheses of previous hydrologic studies create aggregates of benchmark datasets that result in scientific value beyond the original studies, including facilitating robust analyses of science questions at broader temporal and geographic scales than those of their individual efforts (Zhang et al. 2021; Vlah et al. 2023). Such efforts range from general, for example national to global scale collections of streamflow and hydroclimatic data (Caravan [Kratzert et al. 2023]; EStreams [do Nascimento et al. 2024]), to specific, for example a socio-hydrological dataset of paired floods and droughts events (Panta Rhei benchmark dataset [Kreibich et al. 2023]). We posit that synthesizing these data with national data repositories, for example The U.S. Geological Survey National Water Information System (NWIS; U.S. Geological Survey 2024) could support broader analyses of headwater systems.

4.4 | Metadata Considerations

When data are shared, the value and usability of the data can be greatly improved if relevant metadata are also shared.

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Otherwise, as end users access data through repositories and become distanced from the original data, information such as streamflow data uncertainty is lost and becomes unavailable to subsequent analyses (McMillan et al. 2012). Appropriate streamflow metadata remain a persistent challenge and a major limiting factor to data sharing opportunities and continued creation of "dark data" (Ellison 2010; Kaiser, Blasch, and Schmitz 2023). This limitation also prevents the quantification and comparison of uncertainty in streamflow data across multiple datasets. At present, many organizations are collecting streamflow data in headwater streams that are not easy to locate or access. Efforts to create minimum metadata templates may require substantial interaction with the data collection agency, and existing metadata standards can have additional requirements. Additionally, meeting Findable, Accessible, Interoperable, and Reusable (FAIR) data access practices can be challenging because of restrictions in data sharing agreements or data privacy issues (Volk et al. 2014). Work to compile, tag, and integrate existing data and metadata from disparate efforts could increase the usable sample of data available for quantifying and modelling headwater streamflow, but can be a burdensome effort with less data published as a consequence (Kaiser, Blasch, and Schmitz 2023). In an effort to provide guidance toward standardizing hydrologic data publication, CUAHSI includes guidance for standard protocols, publishing to recognized data repositories, creating comprehensive metadata, and promoting interoperability. Similarly, the IoW offers solutions that are flexible with regard to the level of detail included with individual datasets. Efforts like the IoW and others can allow hydrologic data to be used in a myriad of ways. For example, Probability of Streamflow Permanence Model (PROSPER) efforts in the Pacific Northwest and Upper Missouri River basins (Jaeger et al. 2019; Sando et al. 2022; Jiang et al. 2024) have demonstrated that detailed compilation and integration of existing data, including non-federal data, can lead to substantial improvements in modelling capability. Similar compilation efforts of existing data could be used in other parts of the conterminous United States to improve data availability in these regions.

4.5 | Limitations and Assumptions

Our efforts here demonstrate the first attempt to quantify existing streamflow datasets regionally and nationally across the CONUS; however, several assumptions were used to conduct the analyses. First, when aggregating the number of active years, we assumed that the site was active for all years between the start and end period. Thus, there will be an overestimate of active years for sites that were not active for all years within their start and end. For example, it is often the case with USGS stream gages that sites may oscillate between active and inactive status for one or more years at a time, since many of the stream gage sites are funded by and operated for specific data needs of local partners, which fluctuate through time. For determination of stream order, we assumed that while the published coordinates of the sites might not align precisely with the NHD+ HR flowlines (i.e., fall on a flowline), they were located within the correct catchment of the associated flowline. This is generally an acceptable approach (Brinkerhoff et al. 2024; Lane et al. 2026); however, a limited, but unknown number of sites may be associated with the incorrect basin. Our headwater analyses also assume that the NHD is accurate enough to determine headwater status, as defined by Strahler stream order (*sensu* Golden et al. 2025). It has been shown that this is not necessarily true (Christensen et al. 2022). Similarly, data quality is not consistent across these datasets. Further, we do not have a way to summarize and describe the uncertainty associated with these data and, therefore, cannot provide them or a description of them to the public Before these data are used, it is advised that the user inspect each dataset to evaluate if the data are of sufficient quality for their particular purpose. Finally, additional observations and datasets likely exist beyond those that we describe herein, especially where streamflow data may not be publicly available and (or) readily accessible.

5 | Conclusions

This study emphasizes how hydrological data can be used to understand and manage water resources, particularly in the context of headwater streams, which are underrepresented in monitoring efforts. Our inventory of 72 primary and secondary datasets reveals a substantial representation of data collected and published by Federal agencies, and only 6.6% and 25% of stream observation sites are in Strahler stream order 1 and 2 catchments, respectively, at 1:24000 scale. Comprehensive observational data can be used to improve accurate modelling of hydrological processes, assess, and predict the effects of climate change. To address these challenges, a multi-faceted approach that includes low-cost monitoring, citizen science, and remote sensing technologies to enhance data collection in headwater regions might be beneficial. Further, the benefit of submitting the datasets (historical and active) with metadata to data repositories is discussed.

Our analysis demonstrates clear geographic disparities in hydrologic monitoring coverage across CONUS. Physiographic provinces in the eastern U.S. and along the west coast exhibit the strongest representation within the monitoring network, whereas much of the central and western U.S. shows notably sparse observation-day density. Additionally, we find that headwater systems are consistently underrepresented across all physiographic provinces: despite comprising roughly 77% of total stream length, headwaters receive only a fraction of the observation effort relative to downstream systems. This imbalance, reflected in substantially lower observation-day densities for headwaters, highlights a critical gap in the national hydrologic monitoring network and underscores the importance of expanding monitoring efforts to better capture the dynamics of these foundational stream systems.

Furthermore, improvement in the integration and accessibility of hydrological data could facilitate improved water-resource management. Establishing robust, but flexible, data-sharing protocols and comprehensive metadata can enhance the utility of existing datasets and mitigate the issue of "dark data." Because of differences in end-user needs, as well as monitoring objectives and resources, flexibility can be used in the requirements for data standardization and submission.

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The authors have nothing to report.

Data Availability Statement

All data used and referenced in this work are publicly available through the links provided in the manuscript and Supporting Information. All raw data compiled for this analysis are available as a U.S. Geological Survey data release (Sando et al. 2026).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Median length of continuous record (years) and mean discrete record (days) classified by headwaters versus downstream monitoring locations for a 25×25 km grid across the conterminous United States (Sando et al. 2026). **Table S1:** State-, regional-, and national-scale hydrologic monitoring datasets and associated metadata from 1857 to 2023 located in the conterminous United States. **Table S2:** Column definitions associated with Table S1. **Table S3:**. Data services and repositories for hydrologic monitoring datasets located in the conterminous United States. **Appendix S1:** Supporting Information.

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